Multi-Scale Analysis of Terrestrial Vegetation Phenology Using Wavelet and Empirical Mode Decomposition Methods

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Abstract

This term project investigated the application of time-frequency analysis methods to terrestrial vegetation phenology. The discrete wavelet transform (DWT), the continuous wavelet transform (CWT), and empirical mode decomposition (EMD/EEMD) were used to analyze vegetation indices obtained from the Moderate Resolution Imaging Spectroradiometer (MODIS) over a 20-year period for three sites in Michigan. For each of the three analysis methods, phenology timing metrics were extracted from the time-frequency domain and compared against the existing MODIS Land Surface Dynamics product (MCD12Q2) as an independent validation reference.

Phenology

- Phenology is the study of periodic events in ecological cycles as well as the timing of these events.
- Examples include migratory patterns, colony development, and emergence of vegetation.
- Phenological timing is sensitive to climate variability where shifts in timing serve as direct indicators of climate change impacts on ecosystems.
- Accurate phenological metrics are critical inputs to carbon cycle models, land surface models, agricultural forecasting, and more.

Environmental Remote Sensing

- Vegetation indices are measured using remote sensing technology.
- Vegetation reflects solar radiation differently for different wavelengths.
- Healthy leaves strongly absorb red light and strongly reflect near-infrared (NIR) light due to leaf cell structure. This contrast is the basis for all vegetation indices.
- This project analyzed the Enhanced Vegetation Index (EVI)

$$EVI = G\frac{NIR - RED}{NIR + C_1 \cdot RED + C_2 \cdot BLUE + L}$$

with gain $G = 2.5$, aerosol resistance coefficients $C_1 = 6$, $C_2 = 7.5$, and canopy background adjustment $L = 1$.

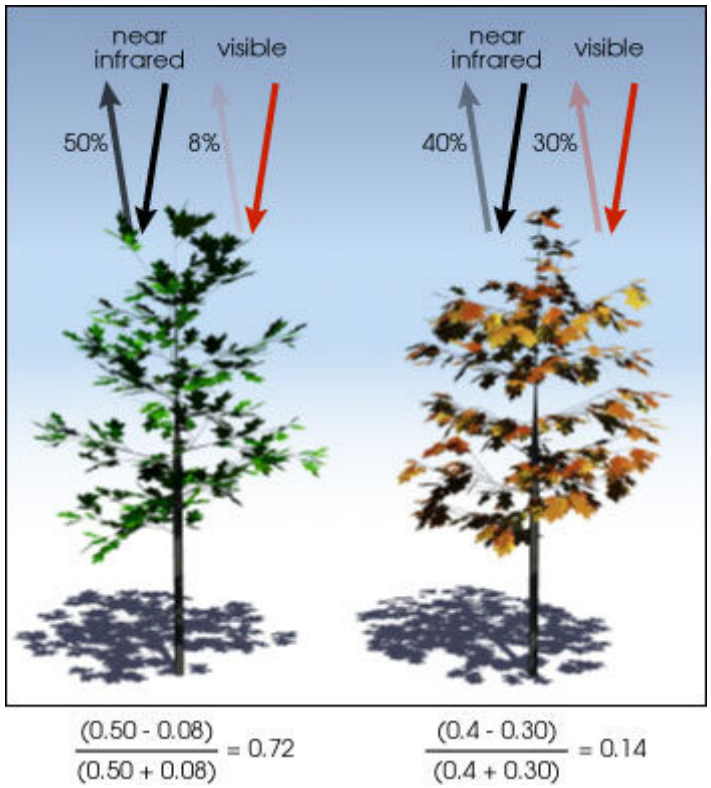


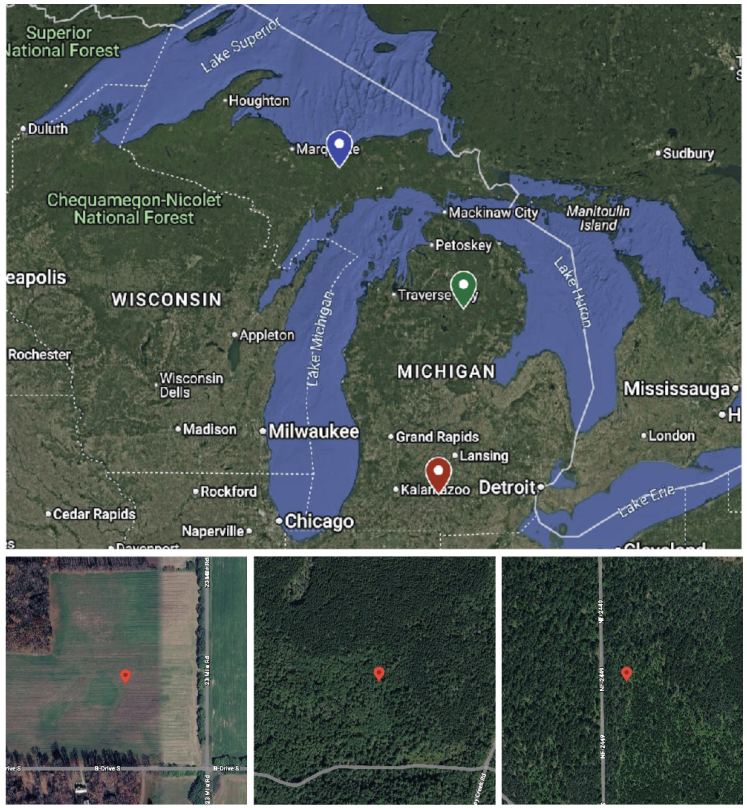
Figure: Image courtesy of NASA.

Problem Statement

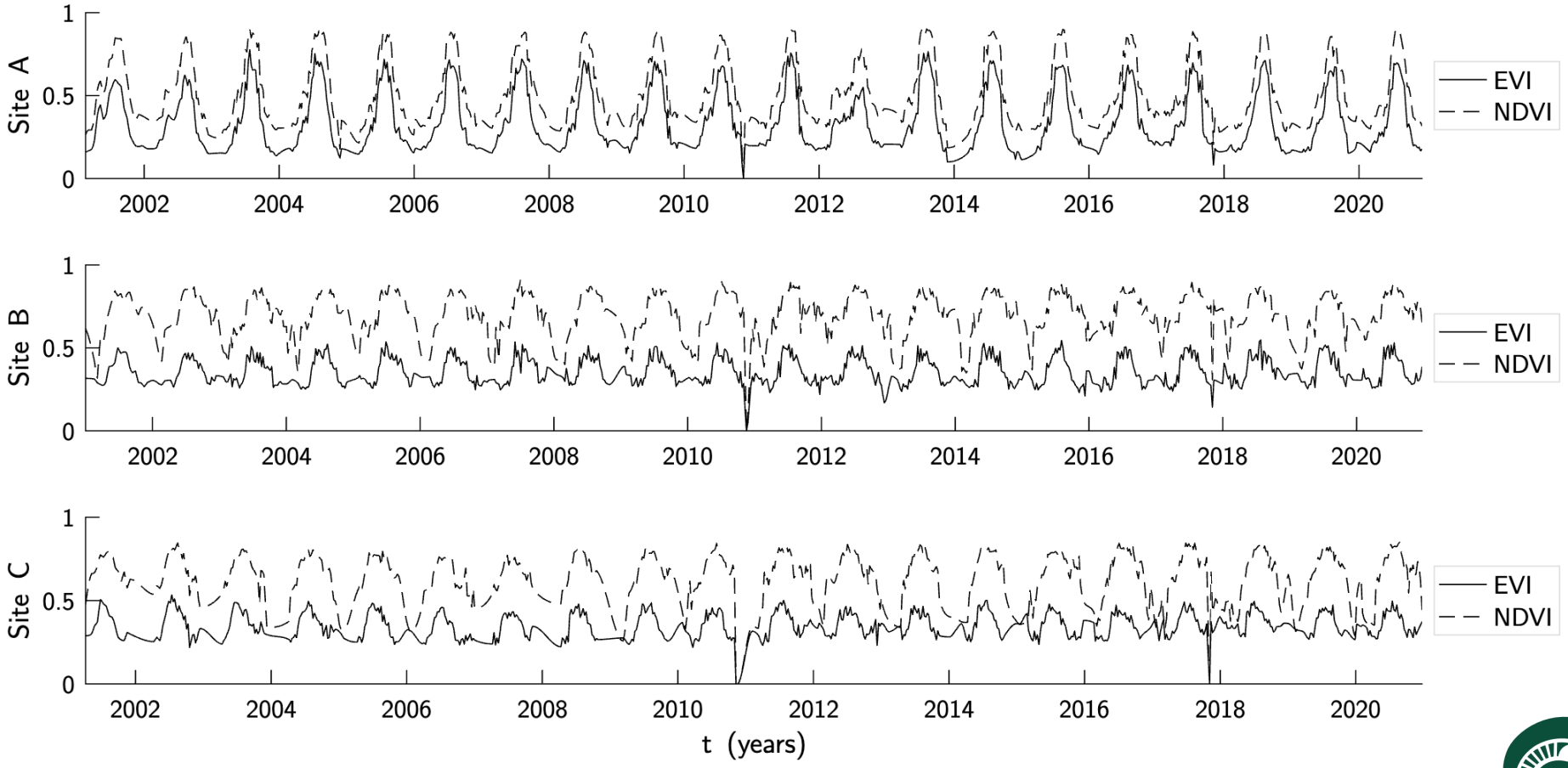
- Classical methods for determining vegetation phenology metrics, such as number of vegetation cycles, the onset of greenness, and senescence, employ threshold-based and curve-fitting approaches on smoothed or reconstructed time series data rather than exploiting the time-frequency structure of these signals.
- This project applied the DWT, the CWT, and EMD to
  1. Demonstrate how phenological signals can be decomposed into sub-seasonal, intra-annual, and inter-annual components using different methods; and
  2. Extract phenological timing metrics directly from the annual scale component extracted from the decomposition.
- Extracted metrics were validated against the MODIS MCD12Q2 Land Surface Dynamics product.

Data

- EVI data was obtained for years 2001-2020 for three sites in Michigan with representation from cropland, mixed deciduous forest, and boreal forest vegetation.
- For each site, a 3-by-3 grid of 500m pixels were extracted. Pixels contaminated by clouds, snow/ice, and shadows were masked out. Next, the pixels were averaged then snapped to an 8-day grid. Finally, missing data values were filled using PCHIP interpolation.
- This presentation reports the results for Site A, the low latitude, cropland site shown by the red marker and in the bottom left satellite view.



Data

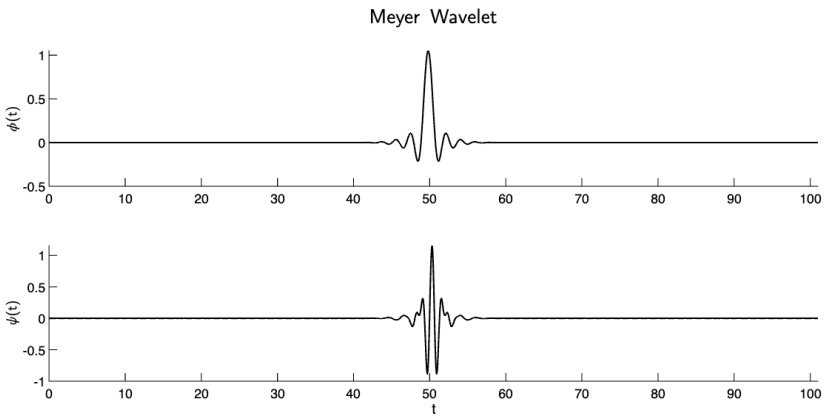

Site A
Site B
Site C
1
0.5
0
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
t (years)
EVI
NDVI

Discrete Wavelet Transform

- The DWT is used to perform MRA on the phenological signals.
- The Meyer wavelet was selected for DWT analysis to align with convention.
- Select the most appropriate scales of study by relating the decomposition level to the period of the reconstructed signals.
- Decomposition level and period can be related by

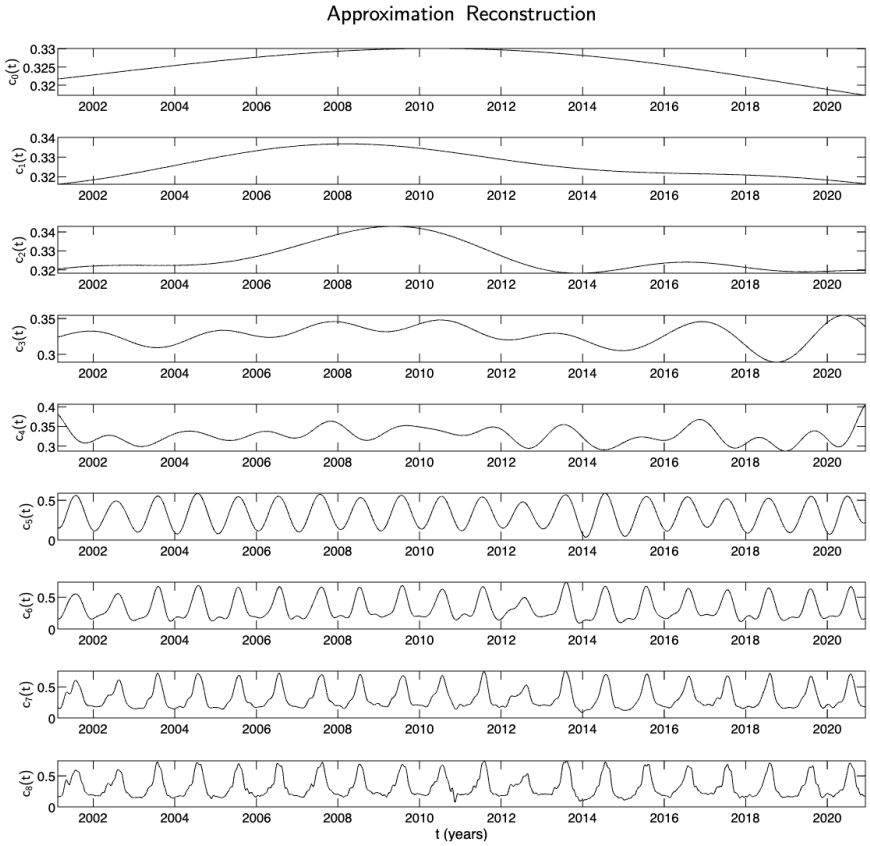
$$p_j = \frac{2^{N-j} T_s}{\nu_c}$$

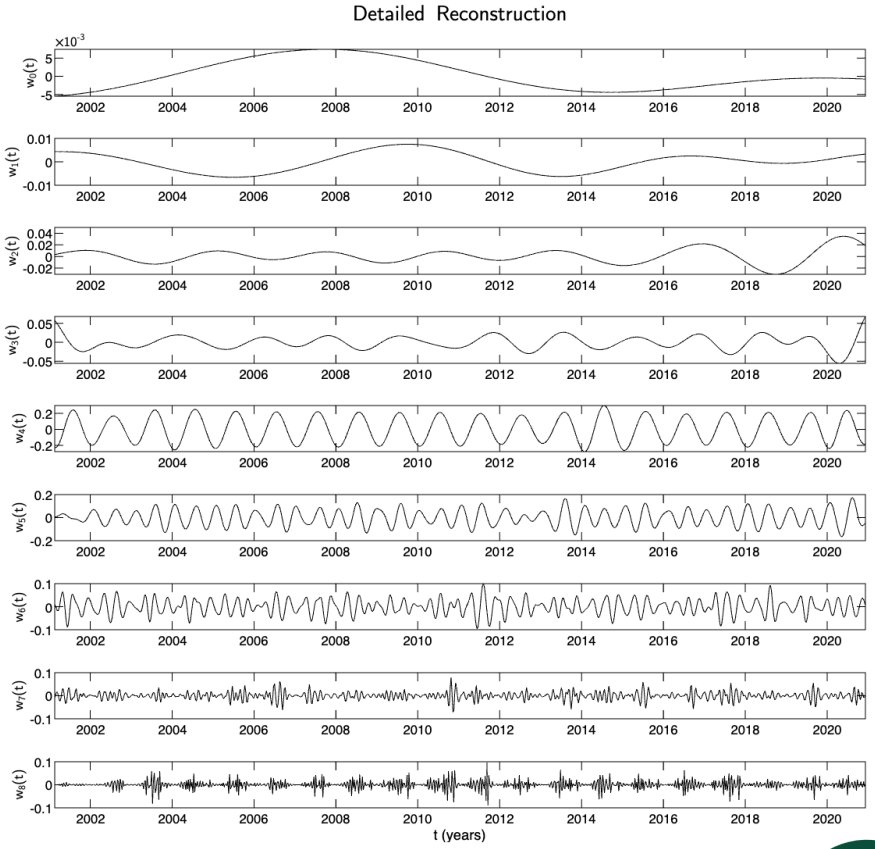
where N is the maximum decomposition level, T_s is the sampling period, and ν_c is the center frequency of the Meyer wavelet.



j	Scale	Period (days)
0	512	6175
1	256	3087
2	128	1544
3	64	772
4	32	386
5	16	193
6	8	96
7	4	48
8	2	24

Discrete Wavelet Transform





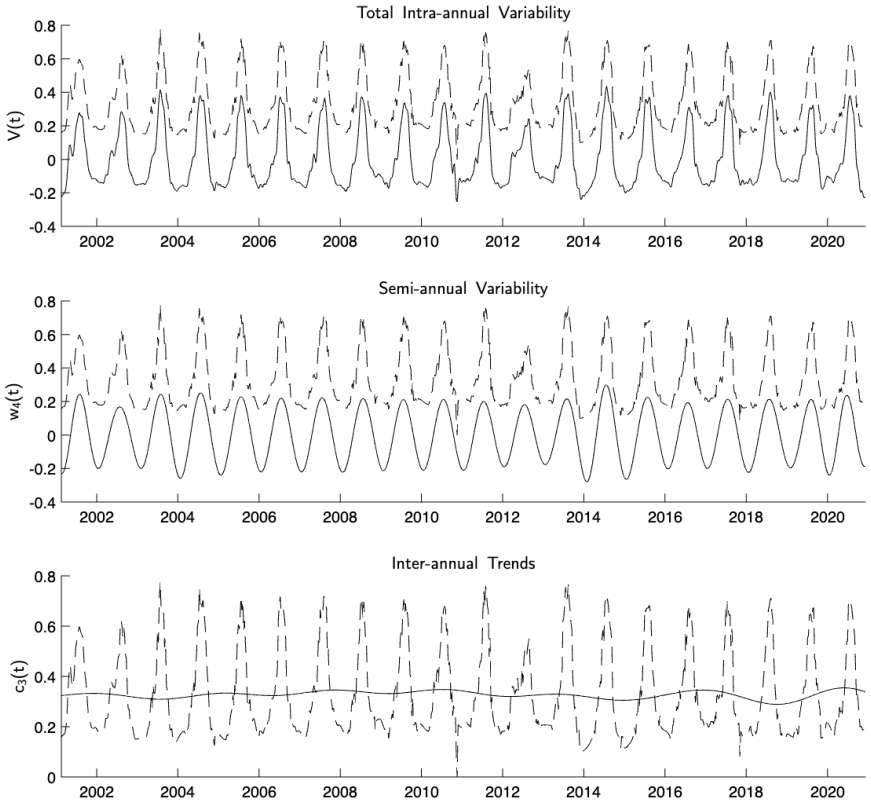
Discrete Wavelet Transform

- The decomposition corresponding to the annual scale is $j = 4$.
- The total intra-annual variability can be captured by

$$V(t) = \sum_{j=4}^{j=7} w_j(t)$$

where $d_8(t)$ is discarded as noise.

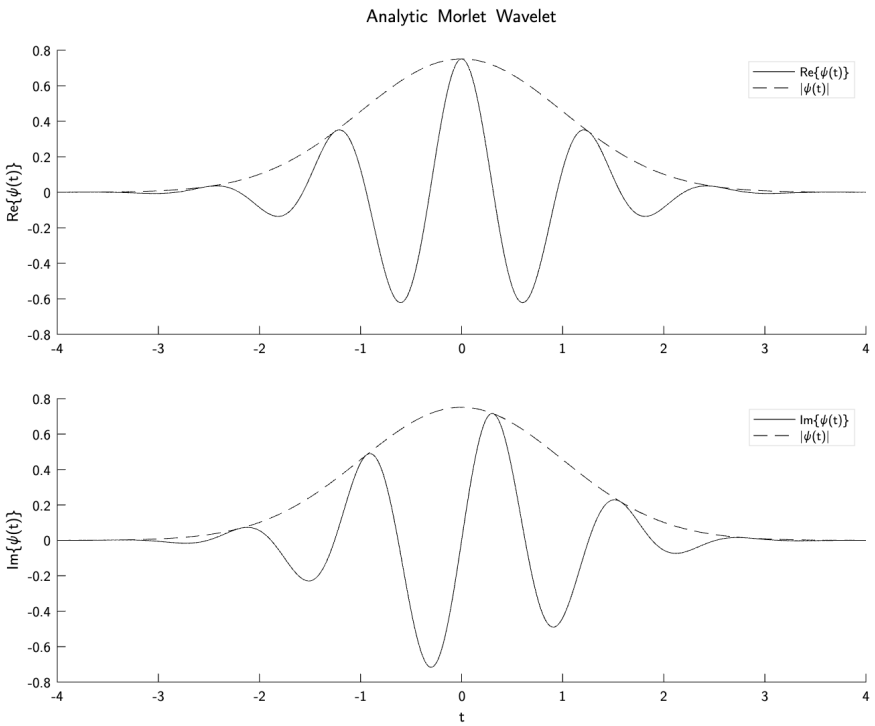
- The semi-annual variability and inter-annual trends can be captured by $w_4(t)$ and $c_3(t)$, respectively.



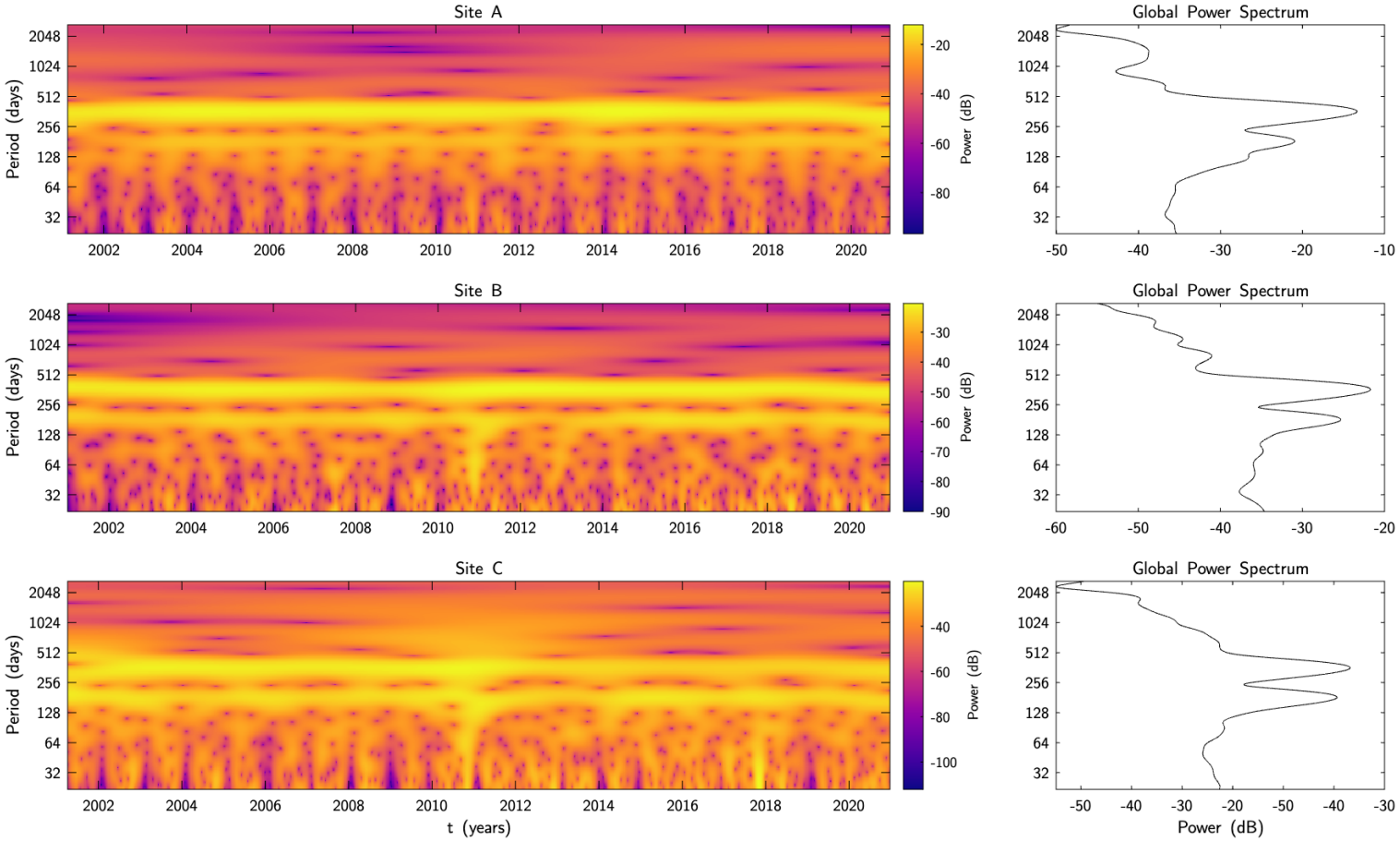
Continuous Wavelet Transform

- The CWT is fundamental to *wavelet coherence analysis*, a technique used to find localized correlations between two signals.
- The Morlet wavelet was selected for CWT analysis to align with convention.
- This wavelet generates a complex-valued transform which is required for wavelet coherence analysis.
- Use band integrated power to capture signal components at different scales.

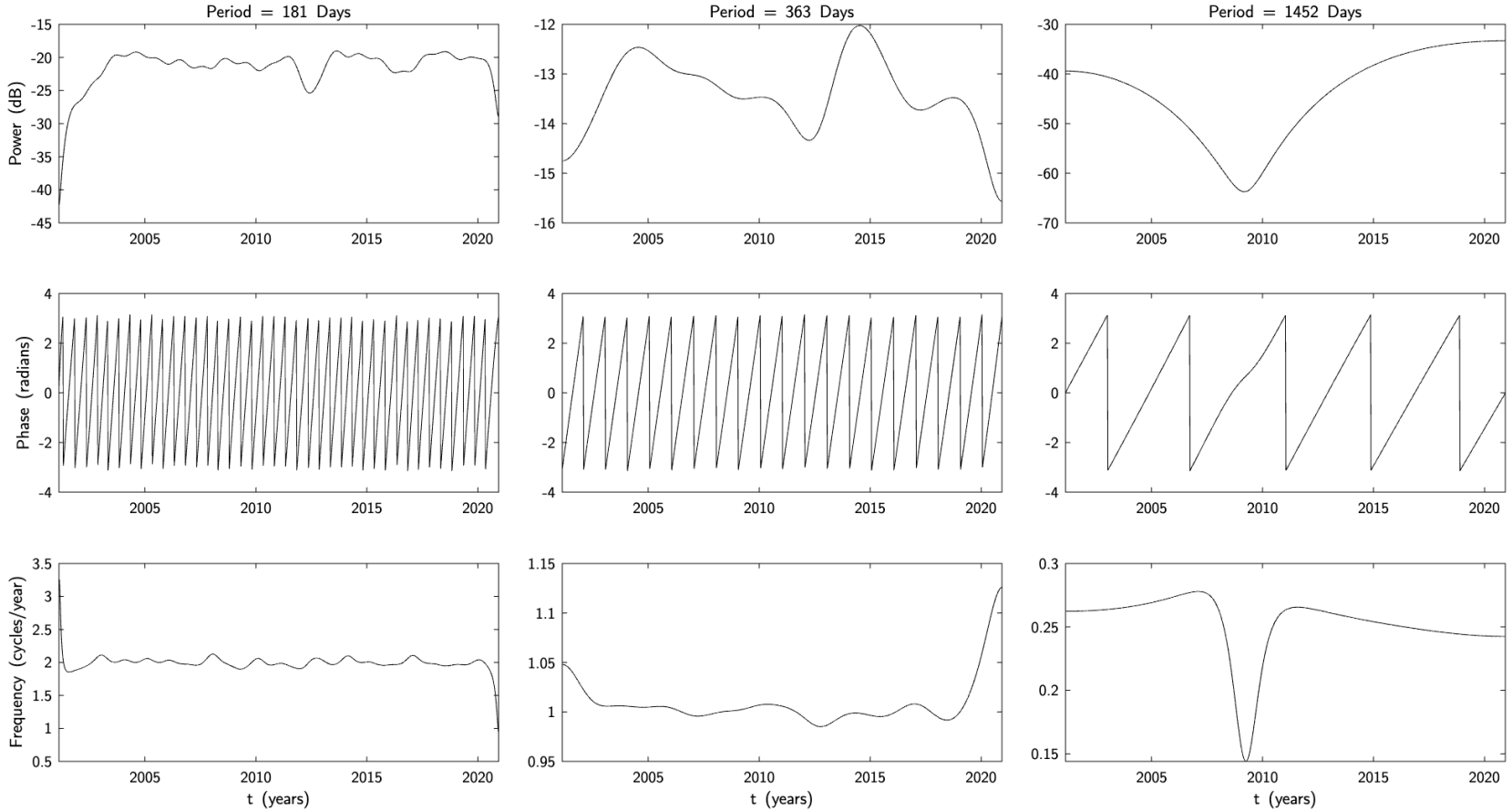
$$P_{band}(t) = \sum_{j:P_j \in [P_1, P_2]} \frac{|W(a_j, t)|^2}{a_j} \delta(\log a)$$



Continuous Wavelet Transform

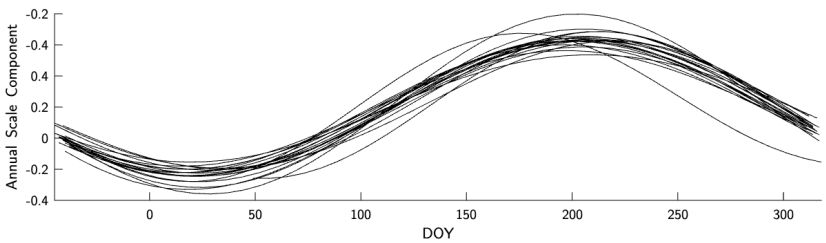


Continuous Wavelet Transform



Timing Metric Extraction

- For each of the three sites, 8 phenological timing metrics were extracted directly from the annual scale component.
- All metrics extracted in phenological year window spanning DOY 320 to DOY 320 to avoid calendar year artifacts that would split the winter dormancy period.
- Thresholds are applied to the amplitude-normalized annual component on each limb independently.

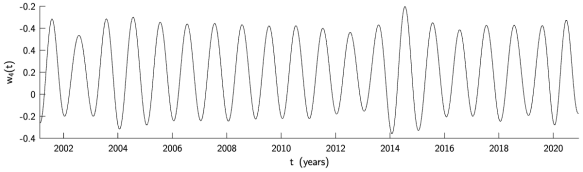


Timing Metric	Definition
No. Cycles	Number of peaks with prominence $\geq$ 10% of maximum value.
Greenup	20% threshold crossing of ascending limb.
Greenup Mid	50% threshold crossing of ascending limb.
Maturity	80% threshold crossing of ascending limb.
Peak	Maximum of annual component.
Senescence	80% threshold crossing of descending limb.
Greendown Mid	50% threshold crossing of descending limb.
Dormancy	Minimum of annual component.

Timing Metric Extraction

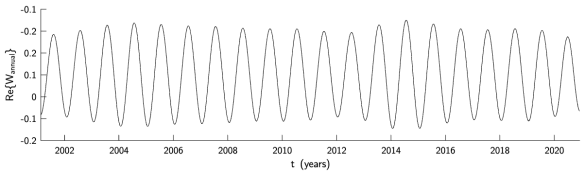
Method 1

- *Transform*: DWT
- Metrics are extracted from from the annual component $w_j(t)$ reconstructed at $j = 4$.



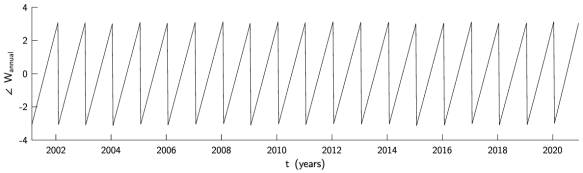
Method 2

- *Transform*: CWT
- Metrics are extracted from the oscillating real part of the wavelet transform $\mathrm{Re}\{W_{annual}\}$.

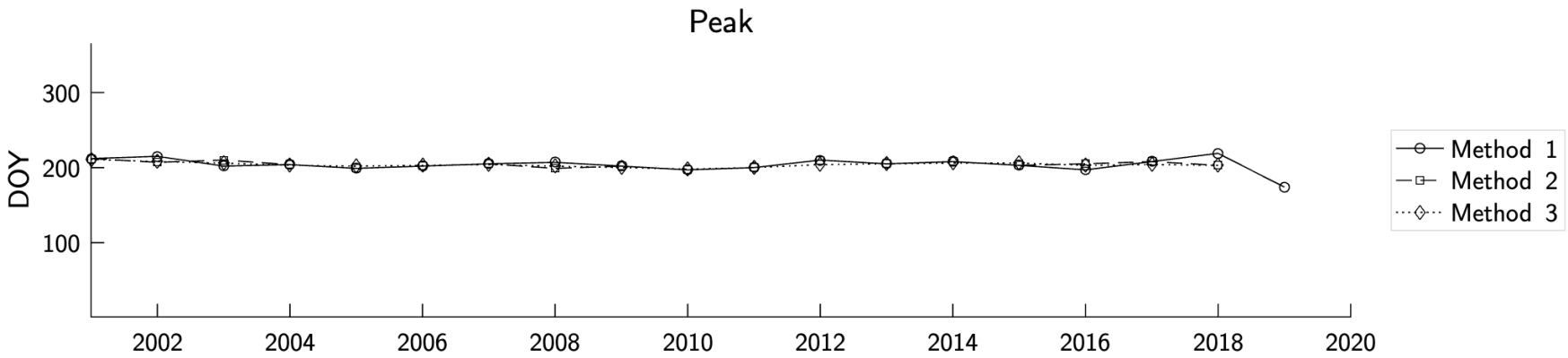


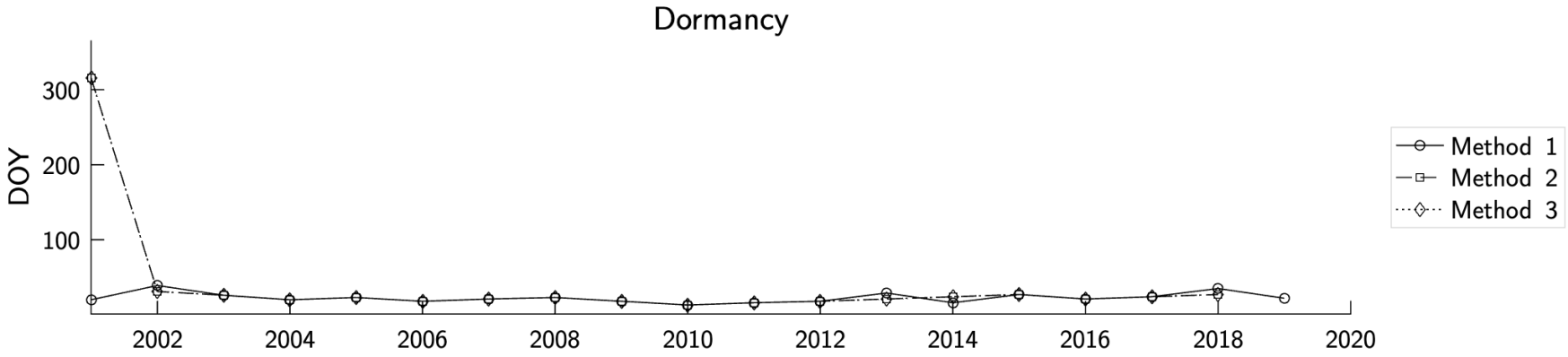
Method 3

- *Transform*: CWT
- Metrics are extracted from the unwrapped phase of the wavelet transform $\angle W_{annual}$ using a reference phase.



Timing Metric Extraction





Evaluation

Method A	Method B	Metric	$\bar{B}$	MAE	$RMSE$	ρ	C_b	ρ_c
1	2	Peak	0.889	2.667	5.333	0.484	0.927	0.448
1	2	Dormancy	4.722	5.611	16.695	0.308	0.630	0.194
1	3	Peak	1.500	3.500	5.083	0.555	0.766	0.425
1	3	Dormancy	4.722	5.611	16.695	0.308	0.630	0.194
2	3	Peak	0.611	1.944	2.506	0.810	0.935	0.758
2	3	Dormancy	0.000	0.000	0.000	1.000	1.000	1.000

$\bar{B}$: Mean Bias

MAE : Mean Absolute Error

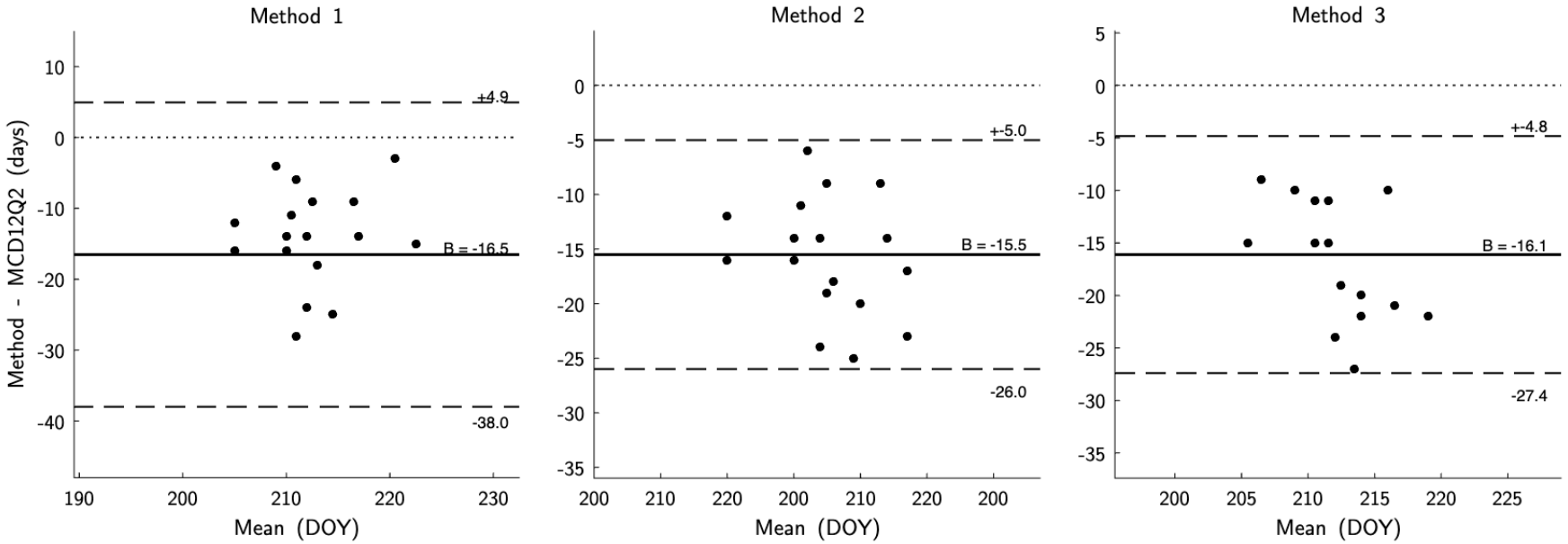
$RMSE$: Root Mean Squared Error

ρ : Pearson Correlation Coefficient

C_b : Bias Correction Factor

ρ_c : Concordance Correlation Coefficient

Evaluation



Method	Metric	$\bar{B}$	*MAE*	*RMSE*
1	Peak	16.526	16.526	19.666
2	Peak	15.500	15.500	16.349
3	Peak	16.111	16.111	17.055

Conclusion

- Each of the DWT and CWT analysis methods successfully decomposed the MODIS EVI time series into physically interpretable sub-seasonal, annual, and inter-annual components.
- Phenological timing metrics extracted from the time-frequency domain are physically coherent and consistent across methods with the three presented methods agreeing on peak greenness timing and dormancy within 2 days and 5 days of each other, respectively. This is suspected to be a result of the differences between the time-frequency localization of the Meyer and Morlet wavelets and is consistent with the time-frequency uncertainty principle.
- All three methods show a consistent 15-16 day negative bias relative to the validation data (MODIS MCD12Q2 product) with the CWT methods showing tighter limits of agreement.
- This work demonstrated that time-frequency domain metric extraction is a viable and theoretically principled alternative to classical threshold-based phenology methods.

Future Work

- Theoretically relate and verify systematic offsets between methods to the time-frequency localization of the wavelet choices.
- Diagnose the source of the 15-16 day bias between the presented methods and the validation data, i.e., is it an artifact of underlying signal representation, interpolation/smoothing, peak/threshold definitions, etc.?
- Explore the choice of different wavelets for each of the DWT and CWT methods and compare against field data to determine which method is most physically meaningful.
- Extend this framework to inform a multi-scale forecasting pipeline for both aquatic and terrestrial vegetation phenology.

Thank you!